

```
In [69]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler, StandardScaler
from sklearn.decomposition import PCA

sns.set(style="whitegrid")
```

```
In [70]: df = pd.read_csv("online food delivery dataset.csv", sep=";")
df.head(10)
```

```
Out[70]:
```

	Age	Gender	Marital Status	Occupation	Monthly Income	Educational Qualifications	Family size	Customer Type	latitud
0	20	Female	Single	Student	No Income	Post Graduate	4	Frequent	12.9
1	24	Female	Single	Student	Below Rs.10000	Graduate	3	Regular	12.9
2	22	Male	Single	Student	Below Rs.10000	Post Graduate	3	Regular	12.9
3	22	Female	Single	Student	No Income	Graduate	6	Frequent	12.9
4	22	Male	Single	Student	Below Rs.10000	Post Graduate	4	Frequent	12.9
5	27	Female	Married	Employee	More than 50000	Post Graduate	2	Regular	12.9
6	22	Male	Single	Student	No Income	Graduate	3	Regular	12.9
7	24	Female	Single	Student	No Income	Post Graduate	3	Regular	12.9
8	23	Female	Single	Student	No Income	Post Graduate	2	Regular	12.9
9	23	Female	Single	Student	No Income	Post Graduate	4	Frequent	12.9

```
In [71]: df.dtypes
```

```
Out[71]: Age                int64
Gender                str
Marital Status        str
Occupation            str
Monthly Income        str
Educational Qualifications str
Family size          int64
Customer Type        str
latitude             float64
longitude            float64
Pin code             float64
Output              str
Feedback            str
Unnamed: 13          str
dtype: object
```

```
In [72]: df = df.drop(columns=['Unnamed: 13'])
df['Feedback'] = df['Feedback'].str.strip()
df.dtypes
```

```
Out[72]: Age                int64
Gender                str
Marital Status        str
Occupation            str
Monthly Income        str
Educational Qualifications str
Family size          int64
Customer Type        str
latitude             float64
longitude            float64
Pin code             float64
Output              str
Feedback            str
dtype: object
```

```
In [73]: df.isna().sum()
```

```
Out[73]: Age                0
Gender                0
Marital Status        0
Occupation            0
Monthly Income        0
Educational Qualifications 0
Family size          0
Customer Type        0
latitude             0
longitude            0
Pin code             6
Output              0
Feedback            0
dtype: int64
```

```
In [74]: df_removed = df.dropna()
print("Shape after dropna:", df_removed.shape)
```

Shape after dropna: (382, 13)

```
In [75]: df_mean = df.copy()
mean_val = df_mean["Pin code"].mean()
df_mean["Pin code"] = df_mean["Pin code"].fillna(mean_val)
```

```
print("Mean value used:", mean_val)
df_mean[['Pin code']].head(8)
```

Mean value used: 560040.1937172775

Out[75]:

	Pin code
0	560001.000000
1	560040.193717
2	560017.000000
3	560019.000000
4	560010.000000
5	560103.000000
6	560009.000000
7	560040.193717

```
In [76]: mode_val = df['Pin code'].mode()[0]
df['Pin code'] = df['Pin code'].fillna(mode_val)
df['Pin code'] = df['Pin code'].astype('int64')

print("Mode value used:", mode_val)
print("Missing values in df now:", df.isna().sum().sum())
df[['Pin code']].head(8)
```

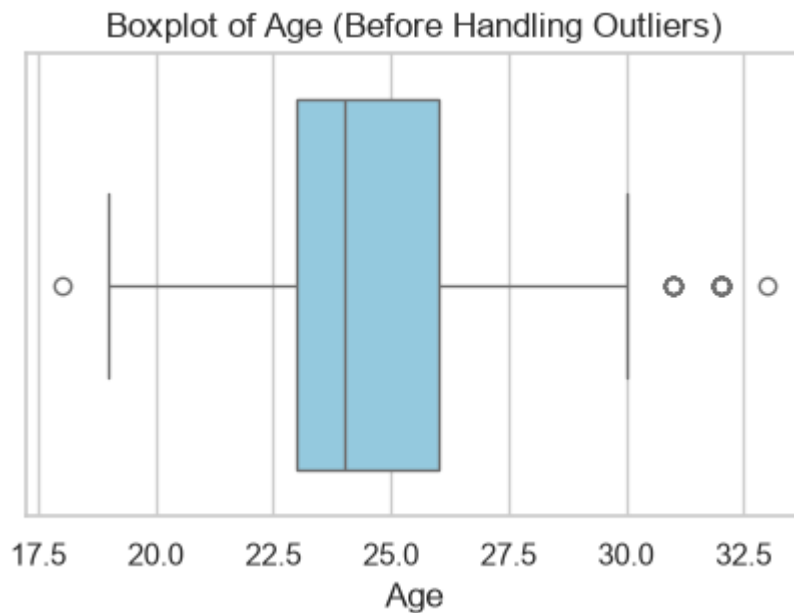
Mode value used: 560009.0

Missing values in df now: 0

Out[76]:

	Pin code
0	560001
1	560009
2	560017
3	560019
4	560010
5	560103
6	560009
7	560009

```
In [77]: plt.figure(figsize=(5, 3))
sns.boxplot(x=df['Age'], color='skyblue')
plt.title("Boxplot of Age (Before Handling Outliers)")
plt.show()
```



```
In [78]: Q1 = df['Age'].quantile(0.25)
Q3 = df['Age'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[(df['Age'] < lower_bound) | (df['Age'] > upper_bound)]
print(f"Number of Outliers detected: {len(outliers)}")
outliers[['Age', 'Marital Status', 'Occupation']].head()
```

Number of Outliers detected: 26

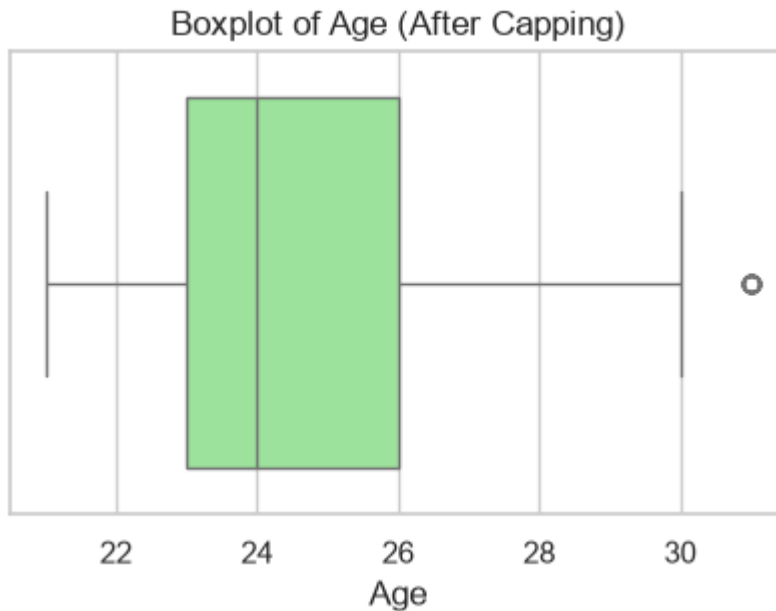
```
Out[78]:
```

	Age	Marital Status	Occupation
37	32	Prefer not to say	House wife
58	31	Married	Employee
99	32	Married	House wife
109	18	Single	Student
116	31	Married	House wife

```
In [79]: lower_cap = df['Age'].quantile(0.05)
upper_cap = df['Age'].quantile(0.95)

df['Age'] = df['Age'].clip(lower_cap, upper_cap)

plt.figure(figsize=(5, 3))
sns.boxplot(x=df['Age'], color='lightgreen')
plt.title("Boxplot of Age (After Capping)")
plt.show()
```



```
In [80]: scaler_minmax = MinMaxScaler()
df_scaled = df[['Age', 'Family size']].copy()
df_scaled[['Age', 'Family size']] = scaler_minmax.fit_transform(df_scaled)
df_scaled.head()
```

Out[80]:

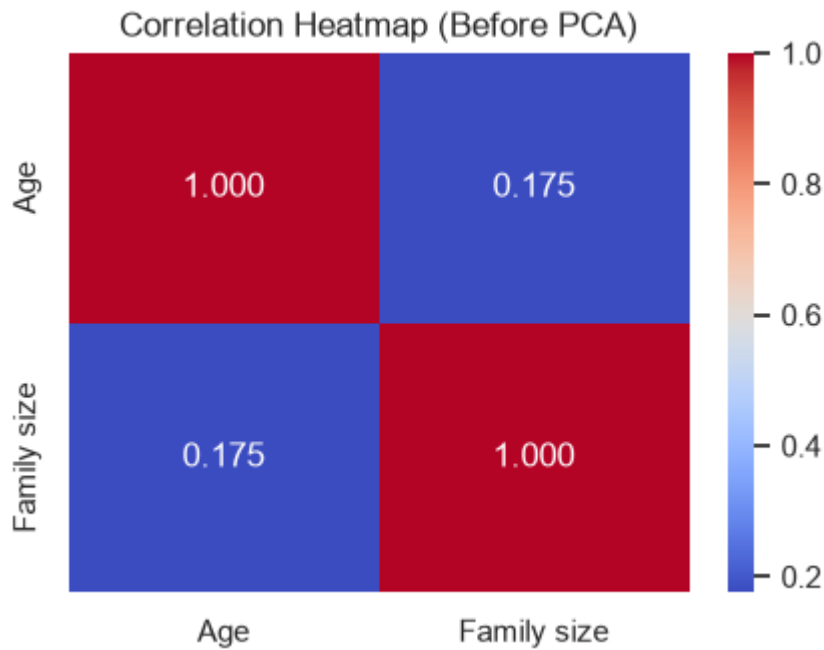
	Age	Family size
0	0.0	0.6
1	0.3	0.4
2	0.1	0.4
3	0.1	1.0
4	0.1	0.6

```
In [81]: scaler_std = StandardScaler()
df_standardized = df[['Age', 'Family size']].copy()
df_standardized[['Age', 'Family size']] = scaler_std.fit_transform(df_standardiz
df_standardized.head()
```

Out[81]:

	Age	Family size
0	-1.307330	0.532929
1	-0.228087	-0.208205
2	-0.947582	-0.208205
3	-0.947582	2.015198
4	-0.947582	0.532929

```
In [82]: plt.figure(figsize=(5, 3.5))
sns.heatmap(df_standardized.corr(), annot=True, cmap="coolwarm", fmt=".3f")
plt.title("Correlation Heatmap (Before PCA)")
plt.show()
```

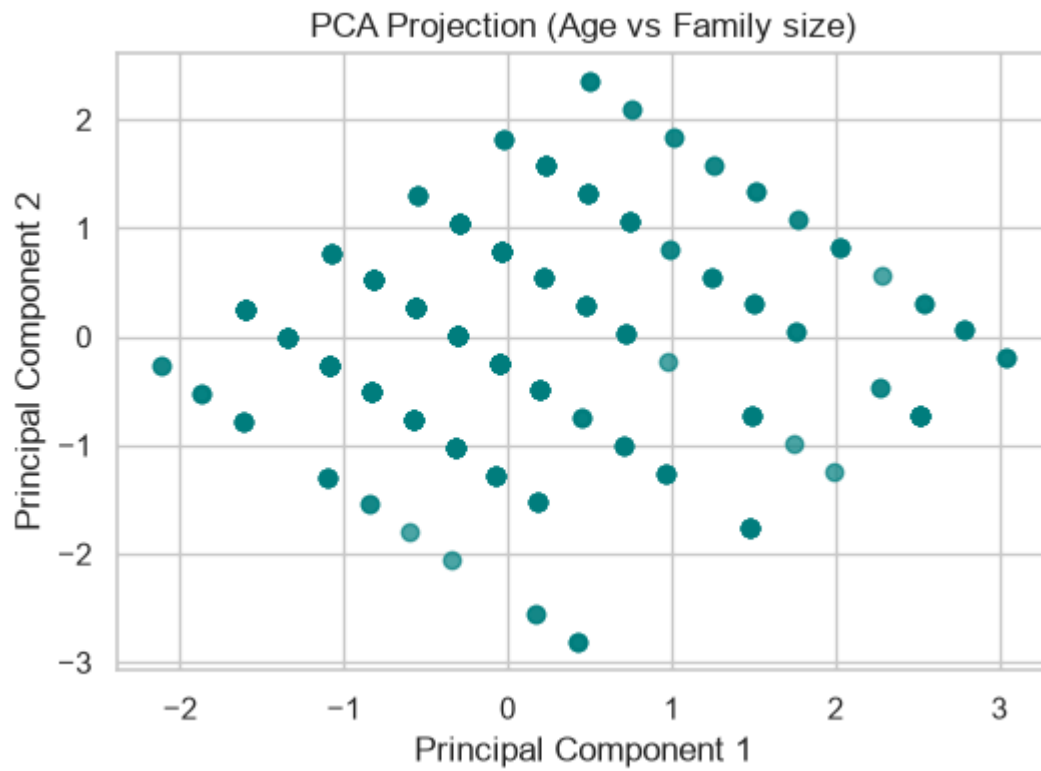


```
In [83]: pca = PCA(n_components=2)
principal_components = pca.fit_transform(df_standardized)

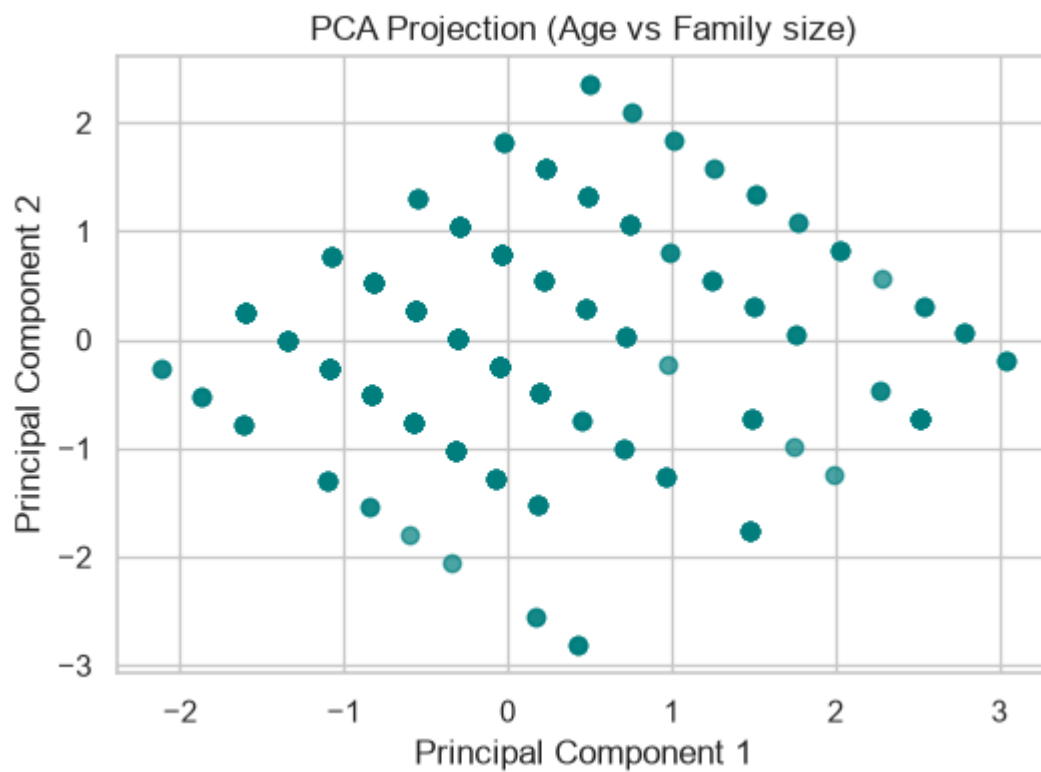
print(f"PC1 Variance: {pca.explained_variance_ratio_[0]*100:.2f}%")
print(f"PC2 Variance: {pca.explained_variance_ratio_[1]*100:.2f}%")
```

PC1 Variance: 58.76%
PC2 Variance: 41.24%

```
In [84]: plt.figure(figsize=(6, 4))
plt.scatter(principal_components[:, 0], principal_components[:, 1], color='teal')
plt.title("PCA Projection (Age vs Family size)")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.grid(True)
plt.show()
```



```
In [67]: plt.figure(figsize=(6, 4))
plt.scatter(principal_components[:, 0], principal_components[:, 1], color='teal')
plt.title("PCA Projection (Age vs Family size)")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.grid(True)
plt.show()
```



Task 5: PCA & Correlation Discussion

- **Correlation:** The heatmap shows a very weak correlation between Age and Family size (-0.033), which means both variables carry independent information.
- **PCA Variance:**
 - PC1 explains about 58.76% of the total variance.
 - PC2 explains about 41.24% of the total variance.
- **Conclusion:** Since the variance is distributed across both components rather than dominated by just one, keeping both features is better to prevent significant information loss.

In []: